



DR ESLAM AHMED  
eliteigcse.com

# Sequences

Find the difference, write the  $n$ th term, and for a series use the sum formula. Every hard question is two equations in  $a$  and  $d$ .

## THE TOPIC ON ONE MAP

### Sequences

#### 01 – Term-to-term and special sequences

next term

$$u_{n+1} = 5u_n - 15$$

Fibonacci-type

$$n^2 + 1$$

#### 02 – The $n$ th term of a linear sequence

38, 31, 24, 17

is 150 a term?

$(2m)$ th term

#### 03 – Arithmetic series: the sum

sum of first 100 terms

41st to 100th term

multiples of 3

#### 04 – Two equations in $a$ and $d$

6th term 39, 19th term 7.8

$$S_{30} = 2880$$

find  $N$

polygon angles

**THE TWO FORMULAS** Every arithmetic question uses one or both — both are on the formula sheet:

#### ZONE TERM

$$u_n = a + (n - 1)d$$

$a$  = first term,  $d$  = common difference

#### ZONE SUM

$$S_n = \frac{n}{2} [2a + (n - 1)d]$$

$$\text{or } \frac{n}{2} (\text{first} + \text{last})$$

# 01 Term-to-Term and Special Sequences

A term-to-term rule makes each term from the one before. Put the term you know into the rule; to go backwards, undo the rule.

## WHEN YOU SEE → YOU DO

**" $u_{n+1} = \sqrt{2u_n + 15}$ ,  $u_1 = 5$ "**

→ substitute:  $u_2 = \sqrt{25} = 5$

**" $u_3 = 6$ , find  $u_2$ "**

→ backwards: solve  $6 = 5u_2 - 15$

**"add the previous two terms"**

→ Fibonacci-type:  $x, y, x + y, \dots$

**" $n$ th term is  $n^2 + 1$ "**

→ put  $n = 1, 2, 3$

**a sequence of fractions**

→  $n$ th term of the top over  $n$ th term of the bottom

## WORKED MODEL • TERM-TO-TERM

$u_{n+1} = 5u_n - 15$  and  $u_3 = 6$ . Find  $u_5$  and  $u_2$ .

**1**  $u_4 = 5(6) - 15 = 15$ ,  $u_5 = 5(15) - 15 = 60$

forwards

**2**  $6 = 5u_2 - 15 \Rightarrow u_2 = \frac{21}{5}$

backwards: solve

**ANSWER**  $u_5 = 60$ ,  $u_2 = 4.2$

## IDEA 01

**Practise: Term-to-Term and Special Sequences**

My questions first, then selected practice.

**Q1** **DR ESAM** Term-to-term

CH 3 • TOPIC 1 2 marks

$u_{n+1} = 2u_n + 3$  and  $u_1 = 4$ . Find  $u_3$ .

---

**SELECTED PRACTICE****Q2** **PRACTICE** Practice • Easy

CH 3 • TOPIC 1 1 mark

4 A sequence is defined using this term-to-term rule.

$$u_{n+1} = \sqrt{2u_n + 15}$$

If  $u_1 = 5$ , find  $u_2$

**(1 mark)**

---

**Q3****PRACTICE**

Practice • Easy

**CH 3 • TOPIC 1****1 mark**

6 These are the first five terms in a sequence.

8   11   14   17   20

Find the next term.

(1 mark)

---

**Q4****PRACTICE**

Practice • Medium

**CH 3 • TOPIC 1****2 marks**

7 The  $n$ th term of a number sequence is  $n^2 + 1$

Write down the first three terms of the sequence.

(2 marks)

---

**Q5****PRACTICE**

Practice • Hard

CH 3 • TOPIC 1

**5 marks**

- 1 (a) The  $n$ th term of a sequence is  $an^2 + bn$ .

Write down an expression, in terms of  $a$  and  $b$ , for the 3rd term.

**(1 mark)**

- (b) The 3rd term of this sequence is 21 and the 6th term is 96.

Find the value of  $a$  and the value of  $b$

You must show all your working.

**(4 marks)**

**Q6****PRACTICE**

Practice • Hard

**CH 3 • TOPIC 1 11 marks**

- 3 (a)** In all the following sequences, after the first two terms, the rule is to add the previous two terms to find the next term.

Write down the next two terms in this sequence.

1   2   3   5   8   13   .....   .....

**(1 mark)**

- (b)** Write down the first two terms of this sequence.

.....   .....   3   11   14

**(2 marks)**

- (c)** (i) Find the value of  $d$  and the value of  $e$ .

2    $d$     $e$    10

**[3]**

- (ii) Find the value of  $x$ , the value of  $y$  and the value of  $z$ .

-33    $x$     $y$     $z$    18

**[5]****(8 marks)**

**Q7****PRACTICE**

Practice • Hard

**CH 3 • TOPIC 1 3 marks**

**13** A sequence is defined by the rule  $u_{n+1} = 5u_n - 15$

If  $u_3 = 6$ , calculate

(i)  $u_5$

$u_5 = \dots\dots\dots [3]$

(ii)  $u_2$

$u_2 = \dots\dots\dots [3]$

**(3 marks)**

## 02 The $n$ th Term of a Linear Sequence

The difference goes in front of  $n$ . Then fix the number: what do you add to reach the first term? To test a number, set the  $n$ th term equal to it — a whole-number  $n$  means yes.

### WHEN YOU SEE → YOU DO

“find the  $n$ th term”, same difference each time

$$\rightarrow dn + (\text{first} - d)$$

a decreasing sequence 38, 31, 24

$$\rightarrow d = -7: 45 - 7n$$

“is 150 a term? Explain”

$$\rightarrow \text{solve } dn + c = 150: \text{whole } n \rightarrow \text{yes}$$

“the  $(n + 1)$ th term” / “the  $(2m)$ th term”

$$\rightarrow \text{replace } n \text{ in the } n\text{th term}$$

“no numbers in both sequences”

$$\rightarrow \text{one is always odd, the other even (or similar)}$$

Difference  $\rightarrow dn$

Fix the number

Check with  $n = 1$

### WORKED MODEL • IS IT A TERM?

2, 5, 8, 11, 14. Find the  $n$ th term. Is 299 a term?

1  $d = 3: 3n$  gives 3, 6, 9  $\rightarrow$  subtract 1

difference first

2  $3n - 1 = 299 \Rightarrow n = 100$

test

ANSWER  $3n - 1$ ; yes, the 100th term

× **TRAP**  $d = -7$  gives  $-7n$ , not  $7n$ . A decreasing sequence has a negative  $n$  term.



## IDEA 02

**Practise: The  $n$ th Term of a Linear Sequence**

My questions first, then selected practice, then real past papers.

**Q8** DR ESLAM  $n$ th term

CH 3 • TOPIC 1 2 marks

Find the  $n$ th term of 5, 9, 13, 17, ... Is 101 a term?

---

**Q9** DR ESLAM Decreasing

CH 3 • TOPIC 1 2 marks

Find the  $n$ th term of 20, 17, 14, 11, ...

---

## SELECTED PRACTICE

**Q10**

PRACTICE

Practice · Easy

CH 3 • TOPIC 1 1 mark

- 1 The first four terms of a sequence are  $-10$   $-8$   $-6$   $-4$

Circle the expression for the  $n$ th term of the sequence.

|            |           |         |           |
|------------|-----------|---------|-----------|
| $-12 - 2n$ | $-8 - 2n$ | $n + 2$ | $2n - 12$ |
|------------|-----------|---------|-----------|

(1 mark)

**Q11**

PRACTICE

Practice · Medium

CH 3 • TOPIC 1 4 marks

- 2 (a) Here are the first four terms of an arithmetic sequence.

$$3 \quad 10 \quad 17 \quad 24$$

Find, in terms of  $n$ , an expression for the  $n$ th term of this arithmetic sequence.

(2 marks)

- (b) Is 150 a term of this sequence?

You must explain how you get your answer.

(2 marks)

**Q12****PRACTICE**

Practice • Medium

CH 3 • TOPIC 1 3 marks

10 (a) Here are the first four terms of an arithmetic sequence.

6      10      14      18

Find an expression, in terms of  $n$ , for the  $n$ th term of this sequence.

(2 marks)

(b) Write down an expression, in terms of  $n$ , for the  $(n + 1)$ th term of this sequence.

(1 mark)

**Q13****PRACTICE**

Practice · Hard

**CH 3 • TOPIC 1 5 marks**

2 (a) Here are the first five terms of an arithmetic sequence.

2   5   8   11   14

Write down an expression, in terms of  $n$ , for the  $n$ th term of this sequence.

**(2 marks)**

(b) Is 299 a term of this sequence?

You must give a reason for your answer.

**(2 marks)**

(c) Write down an expression, in terms of  $n$ , for the  $(n + 1)$ th term of this sequence.

**(1 mark)**

## PAST PAPERS

**Q14** PAST PAPER Edexcel 4MA1 • May 2021 • Paper 1H • Q2

CH 3 • TOPIC 1 2 marks

- 2 Here are the first 4 terms of an arithmetic sequence.

85      79      73      67

Find an expression, in terms of  $n$ , for the  $n$ th term of the sequence.

(Total for Question 2 is 2 marks)

**Q15** PAST PAPER Edexcel 4MA1 • June 2022 • Paper 1H • Q1

CH 3 • TOPIC 1 3 marks

**You must write down all the stages in your working.**

- 1 Here are the first five terms of an arithmetic sequence.

1      5      9      13      17

- (a) Find an expression, in terms of  $n$ , for the  $n$ th term of this sequence.

The  $n$ th term of another arithmetic sequence is  $3n + 5$

- (b) Find an expression, in terms of  $m$ , for the  $(2m)$ th term of this sequence.

(Total for Question 1 is 3 marks)

**Q16**

**PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 1HR • Q2

**CH 3 • TOPIC 1 4 marks**

- 2 In a warehouse there are two types of shelves, type **R** and type **S**.

These two types of shelves are arranged into shelving units that form a sequence of patterns.

Here are the first three terms in the sequence.

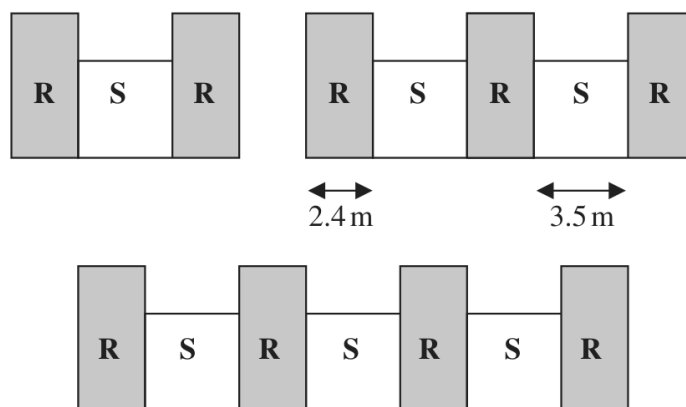


Diagram **NOT**  
accurately drawn

The width of each type **R** shelf is 2.4 m and the width of each type **S** shelf is 3.5 m

- (a) Work out the total width of a shelving unit that has 6 type **R** shelves.

..... m

A shelving unit has  $n$  type **R** shelves.

The total width of this shelving unit is  $W$  metres.

- (b) Find an expression for  $W$  in terms of  $n$   
Give your answer in its simplest form.

$W =$  .....

**(Total for Question 2 is 4 marks)**

**Q17****PAST PAPER**

Edexcel 4MA1 • June 2023 • Paper 2HR • Q4

**CH 3 • TOPIC 1 2 marks**

- 4 Here are the first four terms of an arithmetic sequence.

38      31      24      17

Find an expression, in terms of  $n$ , for the  $n$ th term of the sequence.

.....

(Total for Question 4 is 2 marks)

**Q18****PAST PAPER**

Edexcel 4MA1 • May 2024 • Paper 1H • Q1

**CH 3 • TOPIC 1 3 marks**

**You must write down all the stages in your working.**

- 1 Here are the first four terms of an arithmetic sequence.

1      4      7      10

- (a) Find an expression, in terms of  $n$ , for the  $n$ th term of this sequence.

.....

The  $n$ th term of a different arithmetic sequence is  $5n + 17$

- (b) Find the 12th term of this sequence.

.....

(Total for Question 1 is 3 marks)

## 03 Arithmetic Series: the Sum

Find  $a$ ,  $d$  and how many terms, then use  $S_n$ . For “from the 41st to the 100th”, subtract two sums.

### WHEN YOU SEE → YOU DO

“the sum of the first 100 terms”

$$\rightarrow S_{100} = \frac{100}{2}[2a + 99d]$$

“from the 50th term to the 100th term inclusive”

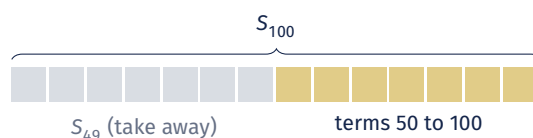
$$\rightarrow S_{100} - S_{49} \text{ (not } S_{50})$$

“sum of multiples of 3 between 1 and 1000”

$$\rightarrow a = 3, d = 3, n = 333$$

“the last term is  $x$ , the sum is 7260”

$$\rightarrow \text{find } n \text{ from the sum, then } x = u_n$$



### WORKED MODEL • MAY/NOV 2020 P1H

8, 15, 22, 29, ... Find the sum of the 50th to the 100th terms inclusive.

1  $S_{100} = 50[16 + 99(7)] = 35\,450$

$$a = 8, d = 7$$

2  $S_{49} = \frac{49}{2}[16 + 48(7)] = 8624$

the first 49

**ANSWER**  $35\,450 - 8624 = 26\,826$



## IDEA 03

**Practise: Arithmetic Series: the Sum**

My questions first, then selected practice, then real past papers.

**Q19** DR ESLAM Sum

CH 3 • TOPIC 1 2 marks

Find the sum of the first 20 terms of 4, 7, 10, ...

---

**SELECTED PRACTICE****Q20** PRACTICE Practice • Medium

CH 3 • TOPIC 1 2 marks

9 Here are the first five terms of an arithmetic sequence.

|   |    |    |    |    |
|---|----|----|----|----|
| 7 | 10 | 13 | 16 | 19 |
|---|----|----|----|----|

Find the sum of the first 100 terms of this sequence.

**(2 marks)**

---

**Q21****PRACTICE**

Practice · Hard

**CH 3 • TOPIC 1****4 marks**

**9** Work out the sum of the multiples of 3 between 1 and 1000  
(4 marks)

**PAST PAPERS****Q22****PAST PAPER**

Edexcel 4MA1 · Oct/Nov 2020 · Paper 1H · Q24

**CH 3 • TOPIC 1****4 marks**

**24** Here are the first five terms of an arithmetic sequence.

8      15      22      29      36

Work out the sum of all the terms from the 50th term to the 100th term inclusive.

(Total for Question 24 is 4 marks)

**Q23****PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 1HR • Q16

**CH 3 • TOPIC 1****4 marks**

**16** An arithmetic series has first term 1 and common difference 4

Find the sum of all terms of the series from the 41st term to the 100th term inclusive.

(Total for Question 16 is 4 marks)

**Q24** **PAST PAPER** Edexcel 4MA1 • November 2023 • Paper 1H • Q19**CH 3 • TOPIC 1** **6 marks**

19 Here are the first 4 terms in an arithmetic sequence.

3      7      11      15

The last term of the sequence is  $x$

The sum of the terms of the sequence is 7260

Find the value of  $x$

Show clear algebraic working.

$x = \dots\dots\dots$

**(Total for Question 19 is 6 marks)**

## 04 Two Equations in $a$ and $d$

Turn each fact into an equation with  $u_n$  or  $S_n$ . Two facts  $\rightarrow$  two equations  $\rightarrow$  solve simultaneously for  $a$  and  $d$ . Then answer the question.

### WHEN YOU SEE $\rightarrow$ YOU DO

“the 6th term is 39, the 19th term is 7.8”  $\rightarrow a + 5d = 39, a + 18d = 7.8$

“three terms  $(2x + 5), (3y - 4), (4x - 2)$ ”  $\rightarrow$  middle  $\times 2 =$  first + last

“ $S_{10}$  is 4 times  $S_5$ ”  $\rightarrow$  write both sums, simplify

“find  $N$ ” / “find  $x$ ”  $\rightarrow$  a quadratic in  $N$ : keep the positive whole answer

“ $u_n > 0$  for all  $n$ ”, “ $d$  is a prime”  $\rightarrow$  use it to reject one solution

Fact 1  $\rightarrow$  equation

Fact 2  $\rightarrow$  equation

Solve for  $a$  and  $d$

Answer

### WORKED MODEL • MAY 2021 P1H

$u_2 = 8.5, u_5 = 13$ , the sum of the first  $N$  terms is 292. Find  $N$ .

1  $a + d = 8.5, a + 4d = 13 \Rightarrow d = 1.5, a = 7$

two equations

2  $\frac{N}{2}[14 + 1.5(N - 1)] = 292 \Rightarrow 3N^2 + 25N - 1168 = 0$

sum

ANSWER  $N = 16$

× **TRAP** Keeping  $N = -\frac{73}{3}$ . The number of terms is a positive whole number.

## IDEA 04

**Practise: Two Equations in  $a$  and  $d$** 

My questions first, then selected practice, then real past papers.

**Q25****DR ESLAM** $a$  and  $d$ 

CH 3 • TOPIC 1

4 marks

An arithmetic series has 4th term 17 and 10th term 41. Find the sum of the first 10 terms.

---

**SELECTED PRACTICE****Q26****PRACTICE**

Practice • Very Hard

CH 3 • TOPIC 1

4 marks

4  $(2x + 23)$ ,  $(8x + 2)$  and  $(20x - 52)$  are three consecutive terms of an arithmetic sequence.

Prove that the common difference of the sequence is 12

**(4 marks)**

---

**Q27****PRACTICE**

Practice · Very Hard

CH 3 · TOPIC 1 5 marks

- 6 The sum of the first 48 terms of an arithmetic series is 4 times the sum of the first 36 terms of the same series.

Find the sum of the first 30 terms of this series.

(5 marks)

**Q28****PRACTICE**

Practice · Very Hard

CH 3 · TOPIC 1 4 marks

- 8 The 10<sup>th</sup> term of an arithmetic series,  $S$ , is 66  
The sum of the first 20 terms of  $S$  is 1290

Find the 5<sup>th</sup> term of  $S$ .

Show your working clearly.

(4 marks)

**Q29****PRACTICE**

Practice • Very Hard

CH 3 • TOPIC 1 5 marks

- 9 The 3rd term of an arithmetic series,  $A$ , is 19  
The sum of the first 10 terms of  $A$  is 290  
Find the 10th term of  $A$ .

(5 marks)

**PAST PAPERS****Q30****PAST PAPER**

Edexcel 4MA1 • January 2020 • Paper 2HR • Q22

CH 3 • TOPIC 1 4 marks

- 22 The first term of an arithmetic series  $S$  is  $-6$   
The sum of the first 30 terms of  $S$  is 2865  
Find the 9th term of  $S$ .

(Total for Question 22 is 4 marks)



**Q31****PAST PAPER**

Edexcel 4MA1 · January 2020 · Paper 2H · Q25

**CH 3 · TOPIC 1 3 marks**

25 Mario is going to save \$50 in the year 2021

He is going to continue to save, up to and including the year 2070, by increasing the amount he saves each year by \$ $k$

Mario will save a total of \$33 125 from 2021 to 2070

Work out the value of  $k$ .

$k = \dots\dots\dots$

(Total for Question 25 is 3 marks)

**Q32****PAST PAPER**

Edexcel 4MA1 · Oct/Nov 2020 · Paper 1HR · Q18

**CH 3 · TOPIC 1 4 marks**

18 In an arithmetic series, the 6th term is 39

In the same arithmetic series, the 19th term is 7.8

Work out the sum of the first 25 terms of the arithmetic series.

$\dots\dots\dots$

(Total for Question 18 is 4 marks)

**Q33****PAST PAPER**

Edexcel 4MA1 • January 2021 • Paper 1HR • Q21

**CH 3 • TOPIC 1 6 marks**

- 21** The  $n$ th term of an arithmetic series is  $u_n$  where  $u_n > 0$  for all  $n$   
The sum to  $n$  terms of the series is  $S_n$

Given that  $u_4 = 6$  and that  $S_{11} = (u_6)^2 + 18$

find the value of  $u_{20}$

.....

**(Total for Question 21 is 6 marks)**

**Q34** **PAST PAPER** Edexcel 4MA1 • January 2021 • Paper 1H • Q24**CH 3 • TOPIC 1** **4 marks**

**24** An arithmetic series has first term  $a$  and common difference  $d$ .

The sum of the first  $2n$  terms of the series is four times the sum of the first  $n$  terms of the series.

Find an expression for  $a$  in terms of  $d$ .  
Show your working clearly.

$a = \dots\dots\dots$

**(Total for Question 24 is 4 marks)**

**Q35****PAST PAPER**

Edexcel 4MA1 • May 2021 • Paper 1H • Q23

**CH 3 • TOPIC 1****5 marks**

- 23** The sum of the first  $N$  terms of an arithmetic series,  $S$ , is 292  
The 2nd term of  $S$  is 8.5  
The 5th term of  $S$  is 13

Find the value of  $N$ .

Show clear algebraic working.

$N =$  .....

**(Total for Question 23 is 5 marks)**

**Q36** **PAST PAPER** Edexcel 4MA1 • November 2021 • Paper 2H • Q20**CH 3 • TOPIC 1** **5 marks**

20 Here are the first four terms of an arithmetic series.

$$k \quad \frac{3k}{4} \quad \frac{k}{2} \quad \frac{k}{4}$$

Given that the 15th term of the series is  $(90 + 2k)$ ,

calculate the sum of the first 30 terms of the series.

.....

(Total for Question 20 is 5 marks)

**Q37****PAST PAPER**

Edexcel 4MA1 • January 2022 • Paper 2H • Q26

**CH 3 • TOPIC 1****5 marks**

- 26 An arithmetic series has first term  $a$  and common difference  $d$ , where  $d$  is a prime number.

The sum of the first  $n$  terms of the series is  $S_n$  and

$$S_m = 39$$

$$S_{2m} = 320$$

Find the value of  $d$  and the value of  $m$

Show clear algebraic working.

$$d = \dots\dots\dots$$

$$m = \dots\dots\dots$$

(Total for Question 26 is 5 marks)

**Q38****PAST PAPER**

Edexcel 4MA1 • June 2022 • Paper 1HR • Q25

**CH 3 • TOPIC 1****5 marks**

- 25** The sum of the first 10 terms of an arithmetic series is 4 times the sum of the first 5 terms of the same series.

The 8th term of this series is 45

Find the first term of this series.  
Show clear algebraic working.

.....

---

(Total for Question 25 is 5 marks)

---

**Q39****PAST PAPER**

Edexcel 4MA1 • June 2023 • Paper 2HR • Q23

**CH 3 • TOPIC 1 5 marks**

23 Here are the first three terms of an arithmetic sequence.

$$8p \qquad 7p - 3 \qquad 4p + 2$$

The sum of the first  $n$  terms of the sequence is  $-1914$

Work out the value of  $n$

Show your working clearly.

$$n = \dots\dots\dots$$

**(Total for Question 23 is 5 marks)**



**Q40****PAST PAPER**

Edexcel 4MA1 • May 2023 • Paper 1H • Q20

**CH 3 • TOPIC 1 6 marks**

**20** The sum of the first 80 terms of an arithmetic series,  $S$ , is 470

The 75th term of  $S$  is 14.5

The sum of the first  $X$  terms of  $S$  is 171

Work out the value of  $X$

Show your working clearly.

$X =$  .....

**(Total for Question 20 is 6 marks)**

**Q41****PAST PAPER**

Edexcel 4MA1 • May 2024 • Paper 1HR • Q23

**CH 3 • TOPIC 1****4 marks**

23 Here are the first three terms of an arithmetic sequence.

$$(4x-14) \quad , \quad (x+2) \quad , \quad (7x-9)$$

Find, as an integer, the sum of the first 40 terms of the sequence.  
Show clear algebraic working.

.....  
(Total for Question 23 is 4 marks)

**Q42****PAST PAPER**

Edexcel 4MA1 • May 2024 • Paper 1H • Q24

**CH 3 • TOPIC 1****6 marks**

**24** A polygon has  $n$  sides, where  $n > 5$

The interior angles of the polygon form an arithmetic sequence.

The smallest angle of the polygon is  $84^\circ$

The common difference of the sequence is  $4^\circ$

Work out the sum of the interior angles of the polygon.

Show clear algebraic working.

○

(Total for Question 24 is 6 marks)

**Q43****PAST PAPER**

Edexcel 4MA1 • November 2024 • Paper 1H • Q19

**CH 3 • TOPIC 1 5 marks**

**19** An arithmetic series has first term  $a$  and common difference  $d$

The sum of the first 30 terms of the arithmetic series is 4395

The sum of the 10th term and the 20th term is 284

Work out the sum of the first 45 terms of the arithmetic series.

Show clear algebraic working.

.....

**(Total for Question 19 is 5 marks)**

**Q44****PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1HR • Q24

**CH 3 • TOPIC 1   6 marks**

24 The first 3 terms of an arithmetic series are

$$(2x + 5) \quad (3y - 4) \quad (4x - 2)$$

where  $x$  and  $y$  are constants.

The sum of the first 9 terms of the series is 216

Find the value of  $x$  and the value of  $y$

Show clear algebraic working.

$$x = \dots\dots\dots$$

$$y = \dots\dots\dots$$

**(Total for Question 24 is 6 marks)**

**Q45****PAST PAPER**

Edexcel 4MA1 • June 2025 • Paper 1H • Q24

**CH 3 • TOPIC 1 5 marks**

**24** An arithmetic series has 30 terms.

The first term is  $a$

The common difference is  $d$

The 20th term is 123

The sum of the 30 terms is 2880

Work out the value of  $a$  and the value of  $d$

Show clear algebraic working.

 $a = \dots\dots\dots$  $d = \dots\dots\dots$ 

**(Total for Question 24 is 5 marks)**

**Q46****PAST PAPER**

Edexcel 4MA1 • November 2025 • Paper 2H • Q25

**CH 3 • TOPIC 1****5 marks**

- 25** The first term of an arithmetic series is 10  
The 20th term of the series is 86

The sum of the first  $N$  terms of the series is 5194

Work out the value of  $N$   
Show your working clearly.

$N = \dots\dots\dots$

**(Total for Question 25 is 5 marks)**

# ALL The Whole Topic on One Page

Tear this page out before the exam.

## ZONE TERM

$$u_n = a + (n - 1)d$$

## ZONE SUM

$$S_n = \frac{n}{2}[2a + (n - 1)d]$$

- |           |                                          |                                                                      |
|-----------|------------------------------------------|----------------------------------------------------------------------|
| <b>01</b> | <b>Term-to-term</b>                      | Substitute forwards; solve backwards.                                |
| <b>02</b> | <b><math>n</math>th term</b>             | Difference in front of $n$ , fix the number; test with a whole $n$ . |
| <b>03</b> | <b>Sum</b>                               | $S_n$ ; a middle block is $S_{\text{last}} - S_{\text{first}-1}$ .   |
| <b>04</b> | <b><math>a</math> and <math>d</math></b> | Two facts, two equations, solve; reject impossible answers.          |

**BOTTOM LINE** Name  $a$  and  $d$  → one formula per fact → solve → answer what was asked.



# Answers

---

Sequences • for the teacher. Every answer checked with a computer algebra system.

## IDEA 01 • TERM-TO-TERM AND SPECIAL SEQUENCES

---

- |           |                         |                                                                          |
|-----------|-------------------------|--------------------------------------------------------------------------|
| <b>Q1</b> | Dr Eslam • Term-to-term | $u_2 = 11, u_3 = 25$                                                     |
| <b>Q2</b> | Practice • Easy         | $u_2 = 5$                                                                |
| <b>Q3</b> | Practice • Easy         | 23                                                                       |
| <b>Q4</b> | Practice • Medium       | 2, 5, 10                                                                 |
| <b>Q5</b> | Practice • Hard         | (a) $9a + 3b$ (b) $a = 3, b = -2$                                        |
| <b>Q6</b> | Practice • Hard         | (a) 21, 34 (b) -5, 8 (c)(i) $d = 4, e = 6$ (ii) $x = 28, y = -5, z = 23$ |
| <b>Q7</b> | Practice • Hard         | (i) $u_5 = 60$ (ii) $u_2 = 4.2$                                          |

## IDEA 02 • THE $n$ TH TERM OF A LINEAR SEQUENCE

---

- Q8** Dr Eslam •  $n$ th term  $4n + 1$ ;  $4n + 1 = 101$ ,  $n = 25$ : yes
- Q9** Dr Eslam • Decreasing  $23 - 3n$
- Q10** Practice • Easy  $2n - 12$
- Q11** Practice • Medium (a)  $7n - 4$  (b)  $7n - 4 = 150$ ,  $n = 22$ : yes
- Q12** Practice • Medium (a)  $4n + 2$  (b)  $4n + 6$
- Q13** Practice • Hard (a)  $3n - 1$  (b)  $n = 100$ : yes (c)  $3n + 2$
- Q14** Edexcel 4MA1 • May 2021 • Paper 1H • Q2  $91 - 6n$
- Q15** Edexcel 4MA1 • June 2022 • Paper 1H • Q1 (a)  $4n - 3$  (b)  $6m + 5$
- Q16** Edexcel 4MA1 • June 2022 • Paper 1HR • Q2 (a) 31.9 m (b)  $W = 5.9n - 3.5$
- Q17** Edexcel 4MA1 • June 2023 • Paper 2HR • Q4  $45 - 7n$
- Q18** Edexcel 4MA1 • May 2024 • Paper 1H • Q1 (a)  $3n - 2$  (b) 77

**IDEA 03 • ARITHMETIC SERIES: THE SUM**

- Q19** Dr Eslam • Sum  $10[8 + 19(3)] = 650$
- Q20** Practice • Medium 15 550
- Q21** Practice • Hard  $\frac{333}{2}(3 + 999) = 166\,833$
- Q22** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1H • Q24  $S_{100} - S_{49} = 26\,826$
- Q23** Edexcel 4MA1 • January 2022 • Paper 1HR • Q16  $S_{100} - S_{40} = 16\,740$
- Q24** Edexcel 4MA1 • November 2023 • Paper 1H • Q19  $n = 60: x = 239$

**IDEA 04 • TWO EQUATIONS IN  $a$  AND  $d$**

- Q25** Dr Eslam •  $a$  and  $d$   $d = 4, a = 5: S_{10} = 5(10 + 36) = 230$
- Q26** Practice • Very Hard  $2(8x + 2) = (2x + 23) + (20x - 52) \Rightarrow x = 5.5$ :  
terms 34, 46, 58,  $d = 12$
- Q27** Practice • Very Hard  $a = -14.5d: S_{30} = 15(2a + 29d) = 0$
- Q28** Practice • Very Hard  $u_5 = 81$
- Q29** Practice • Very Hard  $u_{10} = 47$
- Q30** Edexcel 4MA1 • January 2020 • Paper 2HR • Q22  $d = 7: u_9 = 50$
- Q31** Edexcel 4MA1 • January 2020 • Paper 2H • Q25  $k = 25$
- Q32** Edexcel 4MA1 • Oct/Nov 2020 • Paper 1HR • Q18  $d = -2.4, a = 51: S_{25} = 555$
- Q33** Edexcel 4MA1 • January 2021 • Paper 1HR • Q21  $a = d = 1.5 (u_n > 0): u_{20} = 30$
- Q34** Edexcel 4MA1 • January 2021 • Paper 1H • Q24  $a = \frac{d}{2}$
- Q35** Edexcel 4MA1 • May 2021 • Paper 1H • Q23  $N = 16$
- Q36** Edexcel 4MA1 • November 2021 • Paper 2H • Q20  $k = -20: S_{30} = 1575$
- Q37** Edexcel 4MA1 • January 2022 • Paper 2H • Q26  $m^2d = 242: d = 2, m = 11$
- Q38** Edexcel 4MA1 • June 2022 • Paper 1HR • Q25  $a = 3$
- Q39** Edexcel 4MA1 • June 2023 • Paper 2HR • Q23  $p = 4: n = 29$
- Q40** Edexcel 4MA1 • May 2023 • Paper 1H • Q20  $a = -4, d = 0.25: X = 57$
- Q41** Edexcel 4MA1 • May 2024 • Paper 1HR • Q23  $x = 5: 5380$